

AI Calculator · Test Worksheet

Scan one problem at a time: camera over the middle of the box, page flat, good light. Answer key and score sheet on the last page (keep it out of the camera's view).

1. $\frac{3}{4} + 2^5 \times \sin(30^\circ)$

2. Solve for x:
 $2x + 7 = 15$

3. Solve:
 $x^2 - 5x + 6 = 0$

4. Simplify: $\frac{(2x^3y^{-2})^2}{4x^4y}$

5. Find dy/dx at $x = e$:
 $y = x^3 \ln(x)$

6. Evaluate: $\int_0^2 x \cdot e^{x^2} dx$

7. Find ∇f at $(1, 0)$:
 $f(x, y) = x^2y + e^{xy}$

8. A car starts from rest and accelerates at 3.0 m/s^2 for 8.0 s. How far does it travel?

9. A ball is thrown at 20 m/s, 30° above horizontal. Max height? ($g = 9.8 \text{ m/s}^2$)

A) 5.1 m B) 10.2 m C) 20.4 m D) 2.6 m

10. How many moles are in 36.0 g of H_2O ?

11. $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$. How many grams of O_2 react with 4.0 g of H_2 ?

A) 8 g B) 16 g C) 32 g D) 64 g

12. Pick up milk and eggs after practice.

Handwritten round

Copy each problem into its space by hand, in pencil or pen, your normal writing. Same scan rules.

H1. Copy #2: $2x + 7 = 15$

H2. Copy #3: $x^2 - 5x + 6 = 0$

H3. Copy #4 (the fraction with exponents)

H4. Copy #6 (the integral)

H5. Copy #9 with its A–D options

H6. Write in pencil, small: $\sqrt{(144)} + 3! - 2^3$

- Keep each scan at the same height so results compare fairly. Note the Focus number you used.
- Fold the answer page under or keep it away: if the key is in view, Claude may read the answers.

Answer key and score sheet

Result: ✓ right · G = right but shown as BEST GUESS · ✗ wrong. Also note the % sure and anything misread.

#	Correct answer	Result	% sure	Notes / what it misread
1	$67/4 = 16.75$ ($2^5 \times \frac{1}{2} = 16$)			
2	$x = 4$			
3	$x = 2$ or $x = 3$			
4	x^2/y^5 ($= x^2y^{-5}$)			
5	$4e^2 \approx 29.56$ ($3x^2 \ln x + x^2$)			
6	$(e^4 - 1)/2 \approx 26.80$			
7	$\nabla f = \langle 0, 2 \rangle$ ($f_x = 2xy + ye^{xy}$, $f_y = x^2 + xe^{xy}$)			
8	96 m ($\frac{1}{2} \times 3.0 \times 8.0^2$)			
9	CHOICE A, 5.1 m			
10	2.00 mol ($36.0 \div 18.02$)			
11	CHOICE C, 32 g (31.7 g)			
12	NOTHING TO SOLVE			
H1	$x = 4$			
H2	$x = 2$ or 3			
H3	x^2/y^5			
H4	≈ 26.80			
H5	CHOICE A			
H6	10 ($12 + 6 - 8$)			

Totals	Printed (12)	Handwritten (6)
✓ right		
G best guess, right		
✗ wrong		

- **How to read it:** 15+ of 18 right: the camera is good enough. Printed fine but handwriting fails: lighting, distance or your handwriting size. Small exponents (#4, #6, H3, H4) failing while the rest pass: focus/resolution, the case for the OV5640 autofocus.
- For any miss, open **Show last enhanced image** on the preview page and note whether you can read it yourself.